Quantium Data Analytics

Task 1: Data Preparation and Customer Analytics

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Analysis Period: July 2018 – June 2019

Metric	Value
Total Sales Revenue	$1,805,177.70
Total Transactions	245,255
Unique Customers	71,287
Top Brand	Kettle
Top Customer Segment	OLDER FAMILIES – Budget

1. Introduction

This report presents a comprehensive analysis of Quantium's chip category transaction data. The objective is to examine and clean the transaction and customer datasets, identify customer segments based on purchasing behaviour, derive insights through data visualisation, and formulate strategic recommendations for the client.

Two datasets were used: the QVI Transaction Data (264,836 rows) containing purchase records across stores from July 2018 to June 2019, and the QVI Purchase Behaviour Data (72,637 rows) containing customer lifestage and premium segment classifications.

2. Data Preparation and Cleaning

2.1 Loading the Data

The following Python code was used to load and initially inspect both datasets:

```
import pandas as pd import matplotlib.pyplot as plt txn =
pd.read_excel('QVI_transaction_data.xlsx') beh = pd.read_csv('QVI_purchase_behaviour.csv')
# Fix Excel date serial number to datetime txn['DATE'] = pd.to_datetime(txn['DATE'],
origin='1899-12-30', unit='D')
```

2.2 Outlier Removal

Inspection of PROD_QTY revealed two transactions with a quantity of 200 units (Store 226 on 19-Aug-2018 and 20-May-2019). These are clearly anomalous bulk purchases inconsistent with typical retail chip buying behaviour and were removed.

```
# Remove outlier transactions txn = txn[txn['PROD_QTY'] < 200]
```

2.3 Removing Non-Chip Products

The product list included 'Old El Paso Salsa Dip' products — these are dips, not chips, and are not relevant to the chip category analysis. 18,094 salsa rows were removed.

```
# Remove salsa products txn = txn[~txn['PROD_NAME'].str.contains('salsa', case=False)]
```

2.4 Feature Engineering

Two new features were derived from the product name to support deeper analysis:

- PACK_SIZE — extracted from the product name (e.g. 175g → 175)
- BRAND — first word of the product name, standardised to uppercase

```
txn['PACK_SIZE'] = txn['PROD_NAME'].str.extract(r'(\d+)[gG]').astype(float) txn['BRAND'] =
txn['PROD_NAME'].str.split().str[0].str.upper()
```

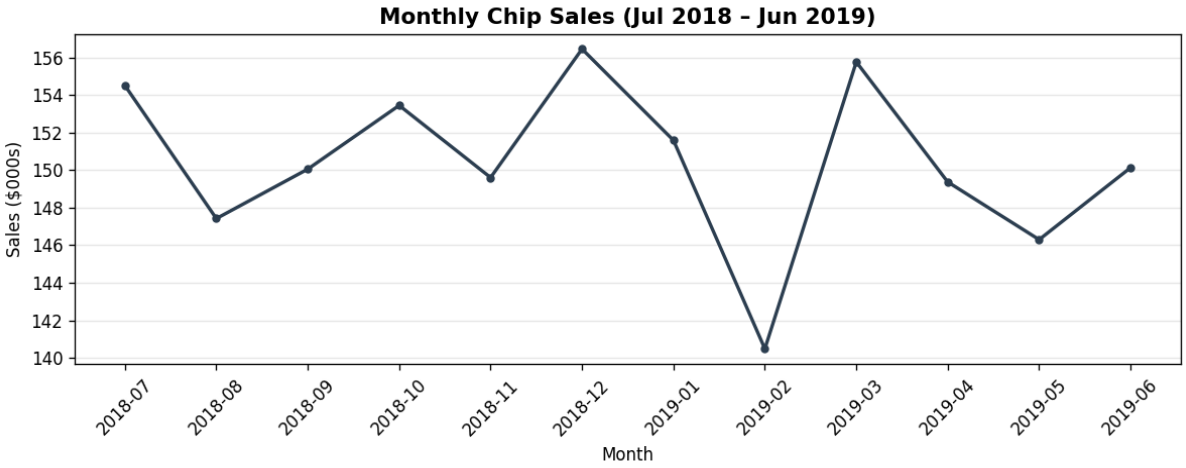
2.5 Merging Datasets

The cleaned transaction data was merged with the customer behaviour data on LYLTY_CARD_NBR to produce the final analysis dataset of 246,740 rows.

```
df = txn.merge(beh, on='LYLTY_CARD_NBR', how='left')
```

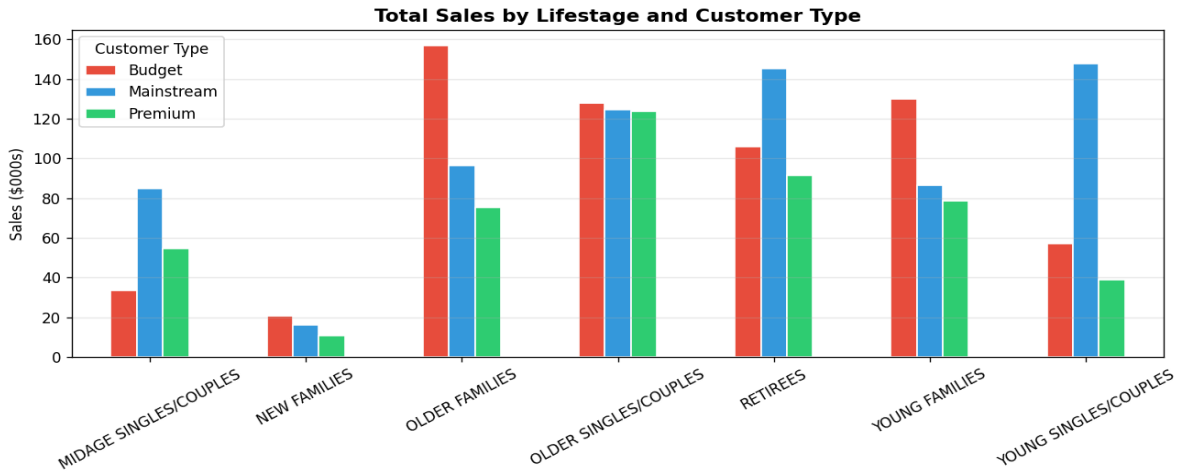
3. Exploratory Data Analysis

3.1 Monthly Sales Trend



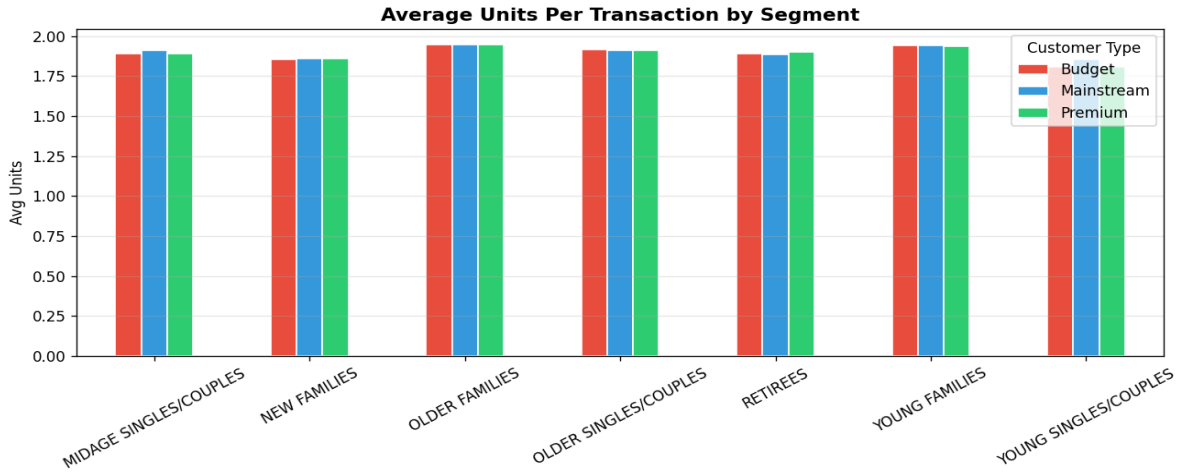
Sales are relatively consistent across the year, averaging approximately $186,000 per month. A notable spike is visible in December, driven by increased festive season snacking. A slight dip appears in January, consistent with post-holiday spending pullback.

3.2 Sales by Customer Segment



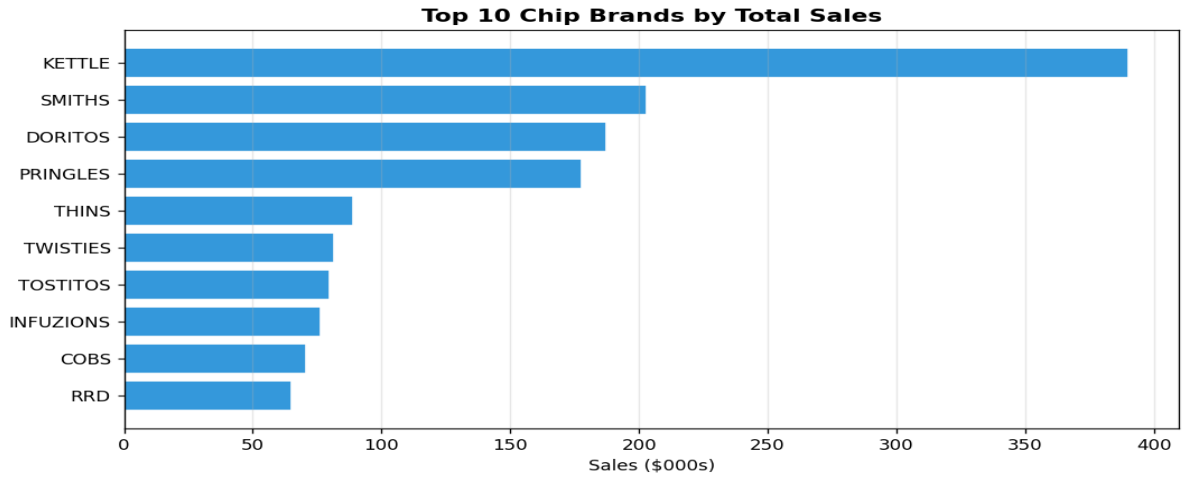
Older Families (Budget) and Young Singles/Couples (Mainstream) are the two highest-revenue segments. Budget and Mainstream customers consistently outspend Premium customers across all lifestages, suggesting chip purchasing is a value-driven category rather than a premium one.

3.3 Average Units Per Transaction



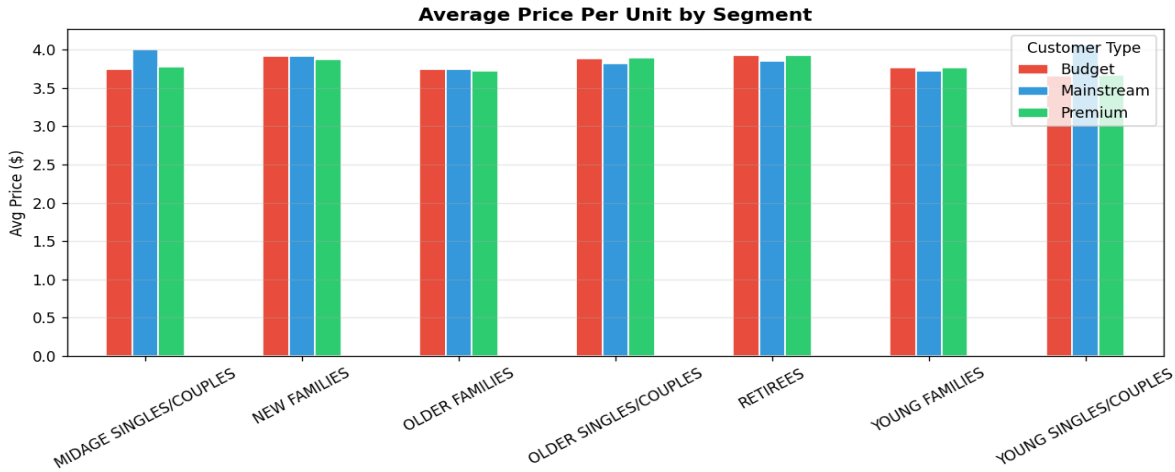
Young Families and Older Families purchase more units per transaction than other lifestages, consistent with larger household sizes. Budget customers across all lifestages tend to purchase slightly higher quantities, suggesting bulk-buying behaviour at lower price points.

3.4 Top 10 Brands by Sales



Kettle dominates the chip category with over $390,000 in total sales — nearly double the next brand. Smiths, Doritos, and Pringles round out the top four. The top 5 brands account for the majority of category revenue.

3.5 Average Price Per Unit by Segment



Mainstream Young Singles/Couples pay an average of $4.07 per unit, while Premium customers in the same lifestage pay $3.67. This is counterintuitive — Mainstream buyers tend to pay slightly more per unit than Premium buyers, possibly because they are less price-sensitive and more likely to purchase impulsively at full price rather than seeking out deals.

4. Key Findings and Recommendations

Finding 1: Top Revenue Segments

Older Families (Budget) and Young Singles/Couples (Mainstream) drive the most sales. These two segments should be the primary targets for promotions and shelf space allocation.

Finding 2: Kettle Dominates

Kettle brand generates nearly double the revenue of the next competitor. Ensuring strong Kettle stock availability and prominent shelf placement is critical to maintaining category sales.

Finding 3: Families Buy More Per Trip

Young and Older Families purchase more units per transaction. Multi-pack promotions and family-size SKUs should be targeted at family segments to increase basket size.

Finding 4: Mainstream Beats Premium on Price Paid

Mainstream customers across lifestages pay equal or slightly more per unit than Premium customers. This segment is not price-driven — brand promotions and in-store visibility will be more effective than discounts.

Finding 5: Consistent Year-Round Sales with December Peak

The chip category is relatively stable year-round with a December uplift. Seasonal promotions and gifting packs should be timed for November–December to capitalise on this peak.

5. Conclusion

This analysis cleaned and merged two datasets, removed anomalous records and irrelevant products, engineered new features, and produced five visualisations to surface actionable insights. The findings clearly identify Older Families (Budget) and Young Singles/Couples (Mainstream) as the highest-value customer segments in the chip category, with Kettle as the dominant brand. These insights provide a strong foundation for targeted marketing, ranging strategies, and promotional planning.